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Roll No

EE/EX-7005(2)-CBGS

B.E. VII Semester

Examination, June 2020

Choice Based Grading System (CBGS)

EHVAC/DC Transmission

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) Discuss the various types of DC links and compare them. 7
b) Draw the schematic diagram of HVDC transmission system and discuss functions of each component. 7
2. a) Compare the EHV AC and DC system of transmission with respect to their power handling capacity. 7
b) Discuss the applications of HVDC transmission system. 7
3. a) Distinguish between 'SVC' and 'STATCOM'. 7
b) Why compensation is required in extra long transmission line? Explain the operation of SVC. 7
4. a) What are the adverse effects of harmonics introduced by HVDC converters? 7
b) Discuss multiterminal DC system (MTDC). Explain it along with advantages and applications. 7

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5. a) How will you achieve control in HVDC system using ignition angle control? Explain it clearly. 7
b) Explain the parallel operation of HVAC and DC system. Also explain its advantages. 7
6. a) Explain the surge impedance of transmission line. Derive its value in terms of line constants. 7
b) What is travelling waves? Discuss how it gets attenuated along the transmission line. 7
7. a) Derive an expression for reflection coefficient of travelling waves at a junction. 7
b) What are the converter faults? Explain protection against over voltages in a converter station. 7
8. Write short notes on the following: 14
 - a) FACTS devices
 - b) Surge Impedance Loading (SIL) and its significance
